

2) Robot Navigation in an Unknown Maze CSA1708

A robot is placed in an unknown maze and must reach the exit without knowing the maze layout beforehand. The robot must explore the environment, avoid obstacles and determine the best path using AI searching techniques.

1. Uniform Cost Search (UCS)

Uniform cost search expands the ~~ex~~ node with the lowest cumulative path cost

Working:

1. Start from the initial position
- 2) store nodes in a priority queue
- 3) Expand the node with the smallest path cost
- 4) Continue until the exit is found.

Advantages:

- Finds the Least cost path
- Complete if path costs are positive
- Optimal solution

Disadvantages:

- High memory usage.
- Slow for very large mazes

Complexity:

Time: Depends on branching factor

Space: Exponential

15
20
B

2. Iterative Deepening Search (IDS)

IDS combines the advantages of Depth First Search (DFS) and Breadth-First Search (BFS)

Working:

1. Search with depth limit 0.
2. Increase depth limit to 1, then 2, and so on.
3. Repeat until the goal is found.

Advantages:

- Very Low memory requirement
- Complete.

D-disadvantages

- Repeats node expansions
- Not suitable for varying path costs.

Complexity:

- Time: $O(b^d)$
- Space: $O(bd)$

Where:

b = branching factor

d = depth of the solution

3. Intelligent Agents in Maze navigation

Model Based Agent

- Stores visited locations
- Builds an internal map.

Goal Based agent

- Plans actions to reach the exit

ex: Smart Home.

- 1) If motion is detected → Turn on the lights
- 2) If No motion is detected
→ Turn off the lights

Advantage

- ~ Fast responses
- ~ Easy to design

Limitation.

- ~ Cannot learn user
- ~ Cannot make decision based on past events

2) Model based reflex agent

- ~ Maintains an internal model of the environment
- ~ Can handle partially observable environment
- ~ Uses current perceived information to make decision. ex: Smart Home

Advantages:

- ~ Better decision making
- ~ Uses memory previous tasks

Limitation:

More complex than simple reflex agent
Require additional memory.

3-Goal based agent

- ~ Makes decision based on achieving a specific goal.
- ~ Uses planning & search techniques.

- Uses search algorithms such as UCS or IDS

4) Best Combination and Justification

Justification:

Goal Based agent focuses on reaching the exit
UCS guarantees the Lowest Cost path

Conclusion:

A Goal Based Agent using UCS provides the most reliable and optimal navigation for an unknown maze.

1) Smart Home Automation: System and artificial intelligence (AI) application that automatically control lights temperature, and security based on user behaviour experimental conditions. The system receives the input from sensors such as motion detectors, temperature, sensors, cameras locks, perform suitable actions to improve comfort, save energy, ensure safety

1. Simple reflex agent.

A Simple reflex agent makes decision based on the current input intercept.

- It follows predefined IF - THEN
- Does not remember previous events

Smart Home.

- ~ Goal: Maintain room temperature 24°C .
- ~ Ensure all doors are locked before midnight

Advantages:

- ~ Intelligent planning
- ~ Can evaluate different actions

Limitation:

- Planning requires more computation
- ~ Slower than reflex agent

4) Utility based agent:

- Select the action that provide the highest overall benefit.

- Balance multiple objectives ex: Smart Home
 - User Comfort
 - Security
 - Electricity Cost

Advantage:

Provide optimal decision

Limitation:

Utility calculations complex

Best agent for Smart Home:

- ~ A Hybrid agent combining model goal utility based agent is the most effective.

Model Based - remember previous states activities goal based - achieves objectives such as maintaining temperature. Ex: Smart home

Utility based $\rightarrow$ Optimizes comfort, energy, efficiency
& security.

